



Fluid structure interaction modeling of liquid sloshing phenomena in flexible tanks

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HIGHLIGHTS

- ▶ We used Ansys Workbench package to study sloshing phenomena in liquid containers.
- ▶ The interaction liquid–structure is modeled considering full and one-way coupling.
- ▶ The results obtained with the FSI models were compared against design codes.
- ▶ The results have shown that the sloshing is influenced by tank wall elasticity.

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ABSTRACT

The present paper is concerned with the problem of modeling the fluid–structure interaction (FSI) in partially filled liquid containers. The study focuses on the sloshing phenomena and on the coupling computational fluid dynamics (CFD) analysis with the finite element stress analysis (FEA) used to predict the sloshing wave amplitude, convective mode frequency, pressure exerted on the walls and the effect of sloshing on the anchoring points forces. The interaction between fluids (water and air) and tank wall is modeled considering full and one-way coupling. Using the time history of an earthquake excitation, the results of the FSI model are compared with those obtained employing simplified mechanical models given in design codes. The coupling phenomenon was found to influence the sloshing effect, the impulsive pressure being amplified by the wall elasticity. The applied FSI methodology proves to be feasible in analyzing a 3D full coupled CFD/FEA storage tank subjected to a long time history excitation.

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1. Introduction

The phenomena of liquid sloshing in storage tanks presents a great interest for several industrial branches such as naval, aerospace, civil, automobile industry and nuclear engineering. For nuclear power plants, the dynamic effects of sloshing could affect the legs and anchor bolts of the storage tanks during strong earthquakes (Shibata et al., 1974). Study of sloshing generally proves challenging due to the presence of strong flow interactions with its container. The primary cause of these greater complexities is the moving boundaries of the fluid as the flexible container deforms under the effect of dynamic sloshing loads (Olsen, 1976; Gradinscak, 2009; Kana, 1982).

Extensive mathematical and empirical relationships have been derived to describe liquid slosh (Ibrahim, 2005; Rebouillat and Liksonov, 2010). These types of analyses are typically undertaken using computational fluid dynamics and finite element methods to solve the fluid–structure interaction problem (Axisa and

Antunes, 2007; Ibrahim, 2005). The fluid flow inside the tank is described by the Navier–Stokes equations, but various assumptions are introduced, such as inviscid and/or incompressible fluid, irrotational motion (Mitra and Sinhamahapatra, 2008; Chen et al., 2009).

From engineering point of view, there are several design codes offering provisions about the seismic behavior of the fluid–structure system (Jaiswal et al., 2003). The procedures recommended in all codes (IITK-GSDMA, 2007; Livaoglu and Dogangun, 2006) use mechanical models (generally a spring mass system) of the fluid–tank system that simplifies the analysis. When a partially filled tank is subjected to an earthquake excitation, the fluid inside the container may be represented dynamically by two uncoupled point masses, respectively impulsive and convective liquid masses (Housner, 1963). This idealized model provides good results only for simple tank geometries, the pressure distribution being given for both rigid and flexible wall tanks (Veletsos, 1984).

Various numerical methods can be used for analysis of liquid storage tanks, among these we mention explicit finite element, implicit Lagrangian–Eulerian, hybrid finite element, Smoothed Particle Hydrodynamics, volume of fluid (Vesjenjak et al., 2004; Lakis et al., 2009; Di Carluccio et al., 2008); a more comprehensive list

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Table 1
Models description.

Model	Method of analysis	Comments
Model I	FSI two-way coupling	FEA–CFD bidirectional coupling
Model II	FSI one-way coupling	CFD–FEA unidirectional coupling
Model III	Spring–lumped masses	FEA dynamic analysis
Model IV	Distributed fluid mass	FEA dynamic analysis

of the 2D and 3D models can be found in Rebouillat and Liksonov (2010). Di Carluccio et al., 2008 have performed a finite element analysis with the LsDyna code for a 3D cylindrical steel tank under a natural earthquake record. The fluid is modeled using a modified structural element defined only by a bulk modulus in order to couple the structure with the fluid through contact elements. Experimental and numerical studies are performed by Jaiswal et al. (2008) for tanks of various shapes other than circular and rectangular shapes and for tanks with internal obstructions. The VOF method firstly proposed by Hirt and Nichols (1981) is used in several papers for modeling the sloshing phenomena. This method is more flexible and efficient than other methods for treating complicated free surface, being able to capture complex surface geometries with overturning or breaking waves and splashing (Eswarana et al., 2009; Ming, 2010). Although numerous studies have been carried out in modeling the sloshing effect, some future research directions are given by Rebouillat and Liksonov (2010), including modeling of 3D elastic tanks subjected to more complex external loads, for example a real earthquake time history.

In this article the coupled and uncoupled sloshing dynamics in a cylindrical vertical ground supported tank were studied using four models developed in Ansys Workbench environment. The main objectives were to verify the feasibility of using a CFD/FEA model for a time-history analysis of a storage tank and to investigate the influence of the coupling upon the sloshing characteristics and reaction loads. The paper is structured as described further. In Section 2, the four models employed are presented. The results of the numerical simulations are given in Section 3, including details of the sloshing characteristics (pressures, forces exerted on walls, wave height, shear forces). Section 4 summarizes the conclusions of the paper.

2. Methods of analysis

Four models were created using the Ansys Workbench package (see Table 1). The first model (Model I) uses the Ansys CFX and Mechanical codes to calculate the wall dynamic pressures in a bi-directional FSI approach. The displacements obtained from the FE code are transferred to the moving boundaries of the fluid domain, and the calculated pressure from CFX is further imposed as a surface load on the liquid–structure interface. The second model (Model II) exports only the solution data from CFX to the mechanical application; the dynamic earthquake effect on the fluid is obtained using a momentum source term. Two simplified mechanical models using

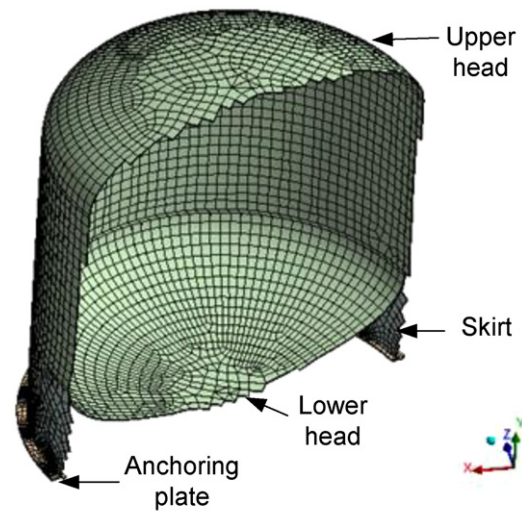


Fig. 1. View of the tank FE model.

lumped (Model III) and distributed (Model IV) fluid mass provide a comparison in terms of base shear force.

The FSI models (Models I and II) take into consideration the following general aspects/simplifications:

- due to lower computational requirements, the two-phase fluid (the liquid and the air/gas in the upper part) motion is considered homogeneous (this assumption is valid only for small amplitudes/smooth sloshing);
- the gaseous phase is taken compressible (this maintains stability because incompressible fluids in a closed domain can react with large pressure spikes to boundary motion);
- no heat transfer is considered in the CFX model;
- the structural domain is meshed using only shell elements;
- the tank is rigidly supported at the base;

2.1. Structural model description

In this study, the typical dimensions of the holding tank are: inner diameter is 3200 mm, thickness of the shell is 10 mm, height is 3294 mm including a skirt of 826 mm high and a base plate of thickness 18 mm. The tank is supported on a skirt and anchored to the plate in concrete by 20 anchor bolts. The tank heads are considered to be torispherical heads with an inner radius of 10% and dish radius equivalent to 80% of the diameter of the tank, the thickness of the heads being 12 mm (Fig. 1). Material data for stainless steel are based on literature (Davis, 1994): density of 7850 kg/m³, Young's modulus of 200 GPa, Poisson's ratio equal to 0.3.

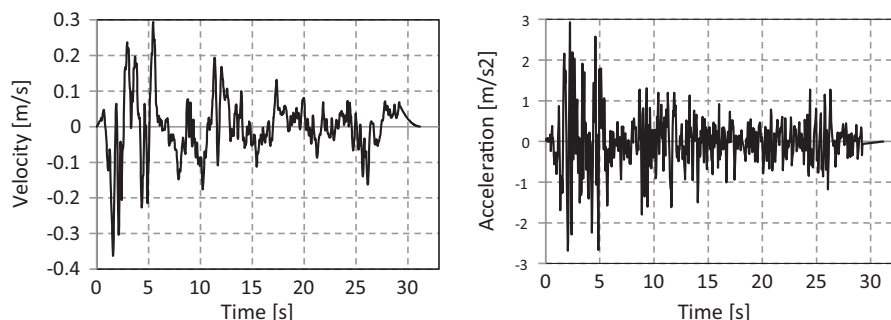


Fig. 2. El Centro Earthquake velocity and acceleration time histories (a – left, b – right).

A finite element model is prepared to represent the holding tank. SHELL181 element is used for modeling the tank wall, tank skirt, and plate. SHELL181 (Ansys Inc., 2009a) is suitable for analyzing thin to moderately-thick shell structures. It is a four-node element with six degrees of freedom at each node.

For the finite element Model I, nodes on the anchor bolts locations have the imposed velocity presented in Fig. 2a representing El Centro earthquake record (May 18, 1940 – Southern California, 6.9 Magnitude). For the second FSI model, the base of the tank is kept as fixed boundary and the equivalent time history acceleration (Fig. 2b) is applied for the entire structure.

For the structural part, the general equation describing the coupled fluid–structure system dynamics can be written:

$$M\ddot{X}(t) + C\dot{X}(t) + KX(t) = -MD\gamma(t) + Q^T p \quad (1)$$

where M , C , K are the system mass, damping and stiffness matrices, D is the ground excitation direction, $\gamma(t)$ denotes the acceleration record, the fluid–structure coupling matrix is represented by Q^T , p is the hydrodynamic pressure and $X(t)$ represents the displacement vector.

2.2. CFD model description

In our study, it was considered that the water level inside the tank is 1.6 m depth and the air above has the pressure of 1 atm. Two-fluid multiphase flows have been employed to model the sloshing dynamics with Ansys CFX code. In homogeneous multiphase flow, a common flow field is shared by all fluids, as well as other relevant fields such as temperature and turbulence. This allows some simplifications to be made to the two-phase model resulting in the homogeneous model. The continuity and momentum equations describing the homogeneous two-phase flow are written:

$$\frac{\partial \rho}{\partial t} + \nabla(\rho \vec{v}) = 0 \quad (2)$$

$$\frac{\partial}{\partial t}(\rho \rightarrow v) + \nabla(\rho \vec{v} \vec{v}) = -\nabla p + \nabla[\mu(\nabla \vec{v} + (\nabla \vec{v})^T)] + S_M \quad (3)$$

in which $\vec{v}$ is the common field velocity, S_M is the user defined momentum source, μ and ρ represents the effective viscosity, density, respectively:

$$\rho(\mu) = \sum r_\alpha \rho_\alpha \quad (4)$$

where r_α and ρ_α denote, respectively, the volume fractions and density of each phase.

In this paper the RANS $k-\varepsilon$ model was employed for the turbulence modeling. Two-equation turbulence models are very widely used, as they offer a good compromise between numerical effort and computational accuracy. In our case we expected smooth sloshing phenomena in which the turbulence effects are well captured by the $k-\varepsilon$ model.

An illustration of the domain mesh is presented in Fig. 3. For a better capture of the free surface interface the mesh was inflated using 10 smooth transition layers in this region. Numerical accuracy and resolution are important issues in CFD calculations. Grids consisting of about 265 000, 116 000 and 50 000 cells were tested for the same case: 1 s transient simulation with a harmonic momentum source. We have observed that the changes for convective and impulsive pressures are less than 5% among the three grid density solutions, and finally we have adopted 116 000 cells as our computational mesh. Because the tank wall was treated as a free slip in the CFD simulation, the grid resolution was not improved near this boundary.

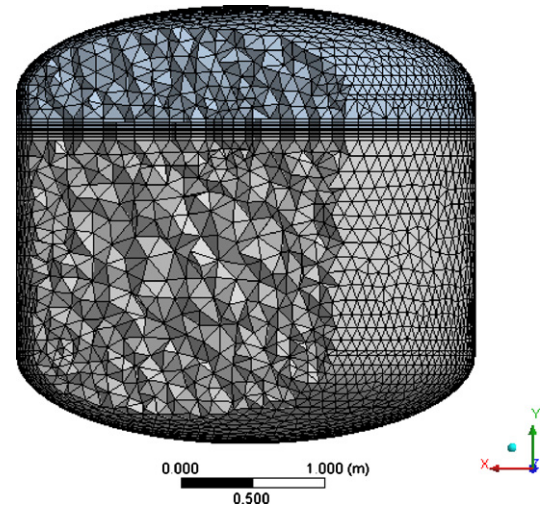


Fig. 3. Fluid domain mesh.

Table 2

Solution methods.

Advection scheme	High resolution
Turbulence numerics	First order
Volume fraction coupling	Coupled
Transient formulation	Second order backward euler
Solver precision	Double

ANSYS CFX uses a coupled solver, which solves the hydrodynamic equations (for u , v , w , p) as a single system. This solution approach uses a fully implicit discretization of the equations at any given time step. The spatial discretization algorithms and the transient formulation are presented in Table 2.

At $t=0$ s the velocity field for the fluid domain is assumed stationary ($\vec{v} = 0$). The reference pressure set in the domain is 1 atm, thus the relative pressure at the free surface level is 0 Pa.

2.3. Coupling procedure

The two solvers (CFX and Mechanical) are linked through Ansys Multi-Field (MFX) capability. During the simulation, the CFX code and the Mechanical application solver execute the simulation through a sequence of multi-field time steps, each of which consists of one or more coupling iterations (Ansys Inc., 2009b). During coupling iterations, each field solver gathers the data it requires from the other solver, and solves its field equations for the current multi-field time step. The details of the procedure used in this paper for bi-directional coupling (FEA $\rightarrow$ CFD) are given in Table 3.

The motion of nodes on the interface boundary is obtained as solution data from the structural application. The motion of all remaining nodes is determined by the mesh motion model, which in Ansys is limited to displacement diffusion. With this model, the displacements in fluid domain are diffused to other mesh points by solving the equation:

$$\nabla(\Gamma_{disp} \nabla \delta) = 0 \quad (5)$$

Table 3

Coupling details.

Stagger Loop Max/Min Iterations	10/3
CFX loop Max/Min iterations	5/2
Under relaxation factor	0.75
Coupling time step	0.04 s

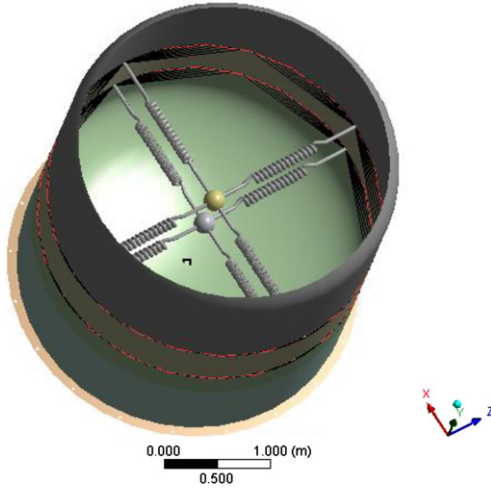


Fig. 4. Spring mass FE model.

In this equation, δ is the displacement relative to the previous mesh locations and Γ_{disp} is the mesh stiffness, which determines the degree to which regions of nodes move together. This equation is solved at the start of each outer iteration or time step for steady state or transient simulations, respectively. The following relation is used to determine the mesh stiffness applied in the displacement diffusion equation:

$$\Gamma_{disp} = \left(\frac{1}{\forall} \right)^{C_{stiff}} \quad (6)$$

where $\forall$ is the control volume size and C_{stiff} is a model exponent ($C_{stiff} = 10$).

For the one-way model the structural domain is fixed in space, but the seismic action upon the fluid is modeled by a time dependent momentum source given by:

$$S_M = \rho \gamma(t) \quad (7)$$

For this model the tank can be considered rigid, the pressures on the walls obtained during the CFD simulation being used in the FEA analysis.

2.4. Simplified mechanical models (Models III and IV)

In order to demonstrate the consistency of the FSI models results and to assess the effect of the free surface motion on the structure comparative with the design codes provisions on seismic analysis of the liquid storage tanks, two simplified mechanical models were developed. As described above, the tank-liquid system can be represented by lumped or distributed fluid mass models, which considerably simplifies the evaluation of hydrodynamics forces.

The spring mass model for ground supported tank is based on work of Housner (1963). Ground supported tanks can be idealized by a spring-mass model shown in Fig. 4. The impulsive mass of liquid m_i is rigidly attached to tank wall at height h_i . Similarly, the convective mass m_c is attached to the tank wall at height h_c by a spring of stiffness K_c (IITK-GSDMA, 2007). This lumped mass model with some variations is almost exclusively used in the specific codes.

The distributed mass model implies that the fluid mass is represented by a modified density tank wall. This means that the total mass of liquid is rigidly attached to the structure. The convective sloshing mode is not considered, thus it is expected to obtain conservative results in terms of base shear in this case.

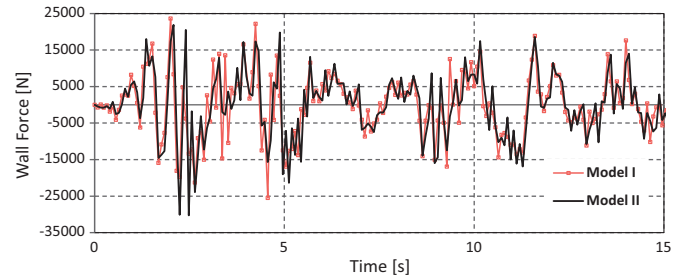


Fig. 5. Wall fluid induced force time variation in x direction.

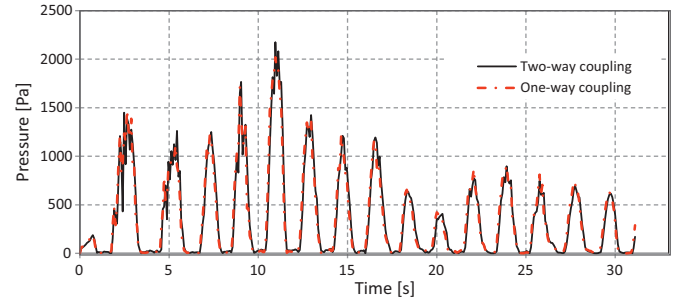


Fig. 6. Convective pressure variation at free surface level.

3. Numerical results

3.1. Comparison between the FSI models and design codes

The total force exerted by the fluid on the tank wall is compared between the models I and II. The differences are observable only in terms of maximum and minimum values, the variation in time being similar for the two cases (Fig. 5). The main component of this integrated pressure is represented by the impulsive hydrodynamic pressure; the ratio of the maximum convective to the maximum impulsive pressures being almost 10% (Figs. 6 and 7). One can observe that the elasticity of the structure has little (almost no) effect on the convective mode, but has a much more influence on the impulsive pressure. This behavior is revealed also by the design codes which express the convective and impulsive frequencies as below (IITK-GSDMA, 2007):

Time period of convective mode for circular tank:

$$T_C = C_c \sqrt{\frac{D}{g}} \quad (8)$$

Time period of impulsive mode for circular tank:

$$T_i = C_i \frac{h\sqrt{\rho}}{\sqrt{t/D\sqrt{E}}} \quad (9)$$

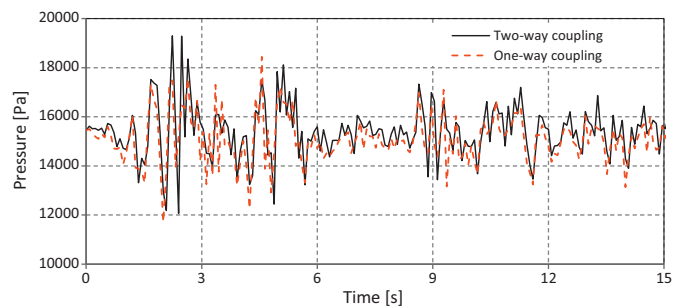


Fig. 7. Impulsive pressure variation at 1.6 m depth.

where $C_c = f(h/D)$, $C_i = f(h/D)$, h is the liquid depth inside the container, D is the inner diameter of tank, t is the thickness of tank wall, ρ represents the liquid density and E is modulus of elasticity of the tank wall.

The C_c and C_i parameters are given as:

$$C_c = \frac{2\pi}{\sqrt{3.68 \tanh(3.68h/D)}} \quad (10)$$

$$C_i = \frac{1}{\sqrt{h/D} \left(0.46 - 0.3h/D + 0.067(h/D)^2 \right)} \quad (11)$$

Calculating using the equations above, we obtain 0.53 Hz for the natural frequency of the convective mode, respectively, 86.2 Hz for the impulsive mode.

Thus, the convective frequency is given only by the geometric characteristics of the tank and liquid depth, while the impulsive mode frequency depends also on the elastic properties of the tank material. Also, from Fig. 6 we can note that during a time period of 31.16 s, the pressure time variation has 16.5 oscillations, corresponding to a frequency of 0.529 Hz, meaning that the sloshing mass is dominated by the fundamental frequency of the convective mode.

An instant of the free surface level is shown in Fig. 8. The chosen time corresponds to maximum wave height reached during the transient (Fig. 9). As we can see, the free surface is smooth, so the assumption of homogenous two phase flow is valid. However, for stronger excitations or for tanks with wave breakers, the inhomogeneous two-phase model have to be used, although is much more computational expensive. The same observation can be made for Fig. 9 as for Fig. 6, e.g. the convective mass dynamics is not influenced by the wall elasticity.

Although the two-way model can provide more realistic results, considering the amplification of the impulsive pressure, one major drawback of the model is the computational time, being almost 50% higher than one-way calculation time (in our case 3.783×10^5 s for Model I and 1.911×10^5 s for Model II).

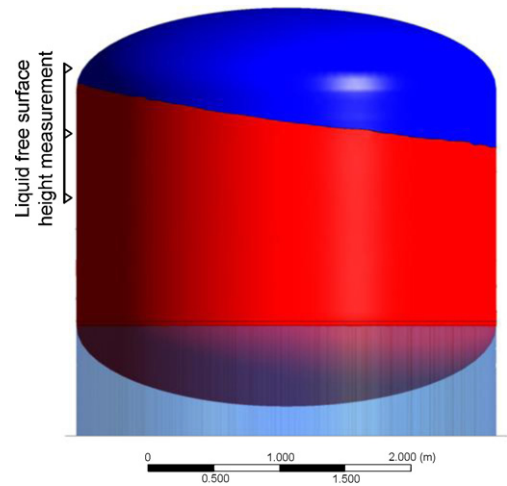


Fig. 8. Liquid level height at 11.04 s.

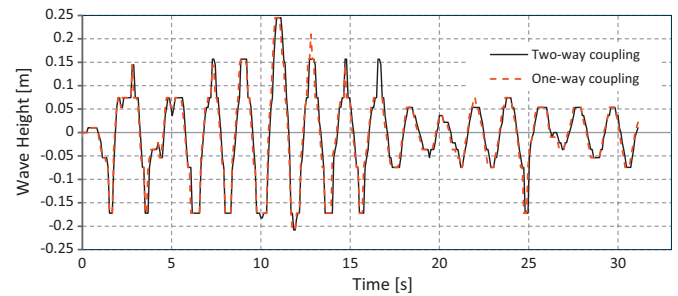


Fig. 9. Free surface oscillations height.

3.2. Anchoring points force evaluation

The time variation of the base shear forces obtained with the four models is presented in Fig. 10. The shear force at the tank base is mainly given by the impulsive component of the fluid, depending

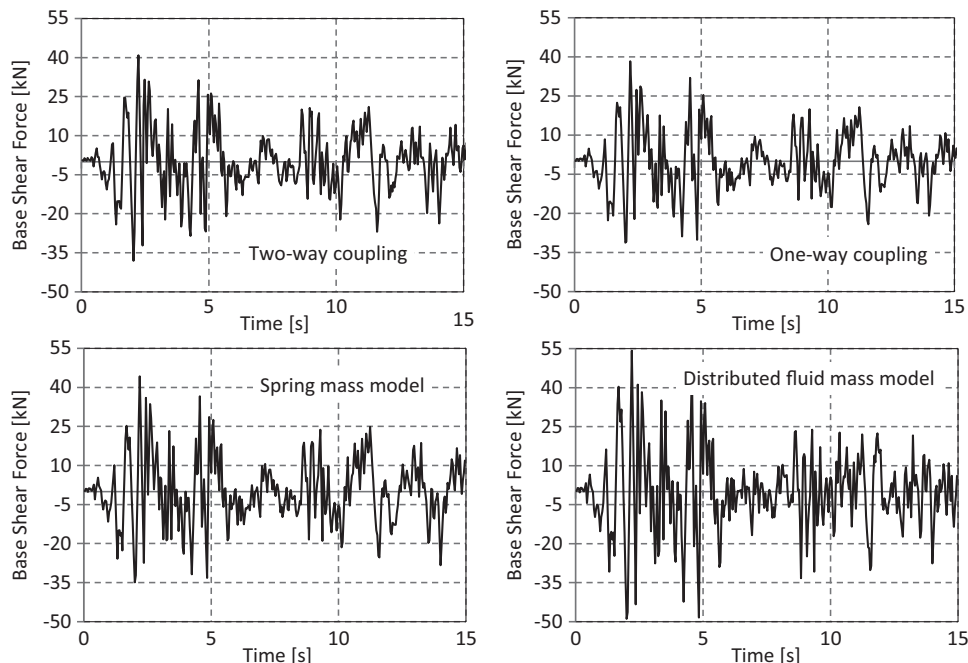


Fig. 10. Shear force in x direction for the four models (excitation direction).

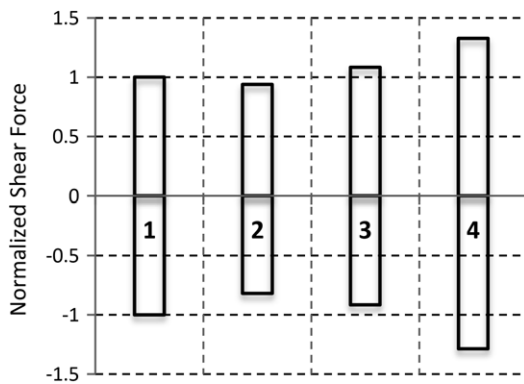


Fig. 11. Comparison of results in terms of maximum shear force (normalized by Model I maximum and minimum values) (1 – Two-way model, 2 – One-way model, 3 – Lumped mass model, 4 – Distributed mass model).

on the wall elasticity. From the point of view of anchoring safety, the maximum and minimum values of the force are important for bolts design. Thus, as we can observe from Fig. 11, the bidirectional FSI and the spring mass models are in good agreement. The one-way coupling model underestimates the base shear force, while the distributed fluid mass model over predicts the maximum shear force. Consequently, these results show that a coupled lumped fluid mass model is more appropriate than the one-way FSI simulation when one wish to assess the reaction force at the anchoring points.

4. Conclusions

Various analytical and numerical models have been proposed in literature to evaluate the non-linear sloshing phenomena, but most of these are limited by tank geometry, input excitation type, tank and support structure flexibility. However, the continuous increase of the computational power and the development of more robust and accurate CFD models allow us to solve the Navier–Stokes set of equations for two-phase flow even for long transients in a reasonable time. Also, several commercial software packages which allow the automatic coupling between FEA and CFD solvers are available.

This paper has focused attention on the advantages of a bidirectional FSI approach in studying the effects of the wall elasticity on the free surface motion of the fluid. The solutions are presented in terms of the impulsive, convective pressure components, liquid wave height and base force reactions. The following conclusions can be drawn from this study:

For complex geometries, the simplified mechanical models are not available for evaluation of reaction base force. The present methodology of coupled CFD and FEA is simple to implement and the results are reliable, showing that the sloshing dynamic loads can be feasible assessed using a bi-directional FSI transient analysis.

Container flexibility modifies the maximum amplitudes of the impulsive pressure; however the liquid slosh height and the convective pressure time history are not dependent on the wall type (considered rigid or elastic).

The coupling phenomena were found to have great significance in the case of fluid–structure interaction analysis. A coupled simplified system (Model III) which takes into account the convective mass provides more accurate results than a one-way FSI uncoupled model (Model II) in terms of base shear force.

The impulsive component of the hydrodynamic pressure is directly proportional to the ground acceleration, both having the same evolution in time.

The convective component can be expressed as a linear combination of sloshing modes with amplitudes dependent on ground acceleration. For smooth sloshing, the first convective mode seems to be dominant (as was the case in our study).

Even though the tank geometry used in this paper is relatively simple, it is expected that the two-way FSI method will give good result for containers with internal structures, even though an inhomogeneous two-phase model may be necessary, which can increase the computational time.

As a final conclusion, the use of a one-way model to analyze the sloshing phenomena in partially filled containers could under predict (the differences are increasing with the tank diameter) the maximum/minimum values of the base shear force, leading in some situations to anchorage failure.

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